Predictive Analytics - Assignment 1 (Solutions)

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```r
library(fpp3)
```

```
## -- Attaching packages ---------------------------------------------- fpp3 0.5 --

## v tibble      3.2.1     v tsibble     1.1.4
## v dplyr       1.1.4     v tsibbledata 0.4.1
## v tidyr       1.3.0     v feasts      0.3.1
## v lubridate   1.9.3     v fable       0.3.3
## v ggplot2     3.4.4     v fabletools  0.3.4

## -- Conflicts ------------------------------------------------- fpp3_conflicts --
## x lubridate::date()    masks base::date()
## x dplyr::filter()      masks stats::filter()
## x tsibble::intersect() masks base::intersect()
## x tsibble::interval()  masks lubridate::interval()
## x dplyr::lag()         masks stats::lag()
## x tsibble::setdiff()   masks base::setdiff()
## x tsibble::union()     masks base::union()
```

Instructions

The data used in this assignment are part of the AER package, which therefore need to be installed and loaded in order to solve this exercise. Relevant documentation about the AER packeage can be found at https://cran.r-project.org/web/packages/AER/AER.pdf. To load the data, type data("DatasetName",package = "AER"). You are supposed to know what are the relevant packages to be used for this assignment.

The deadline for this assignment is the 10 of March 2023. Assignments must be submitted by noon on the Digital Eksamen platform https://eksamen.cbs.dk/, preferably in pdf format. You can work in groups of up to 4 people.

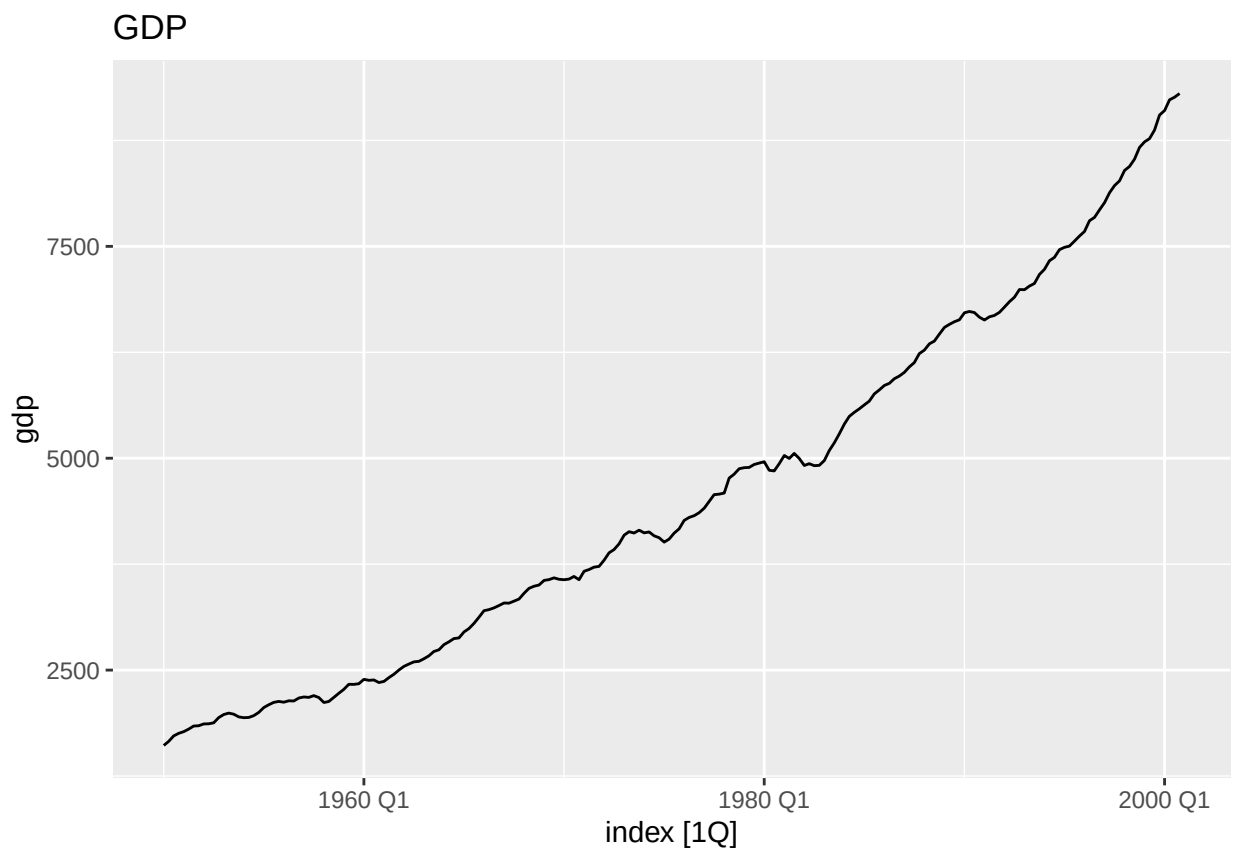
For this assignment, we will study real gross domestic product (GDP from now on) and real government expenditures (public expenditure from now on) in US from 1950 to 2000. The dataset USMacroG contains 12 time series: Real gross domestic product (in billion USD), Real consumption expenditures, Real investment by private sector, Real government expenditures, Real disposable personal income, Consumer price index, Nominal money stock, Quarterly average of month end 90 day treasury bill rate, Unemployment rate, Population (in million), Inflation rate, Ex post real interest rate (computed as treasury bill rate - inflation). Data have quarterly frequency.

Questions

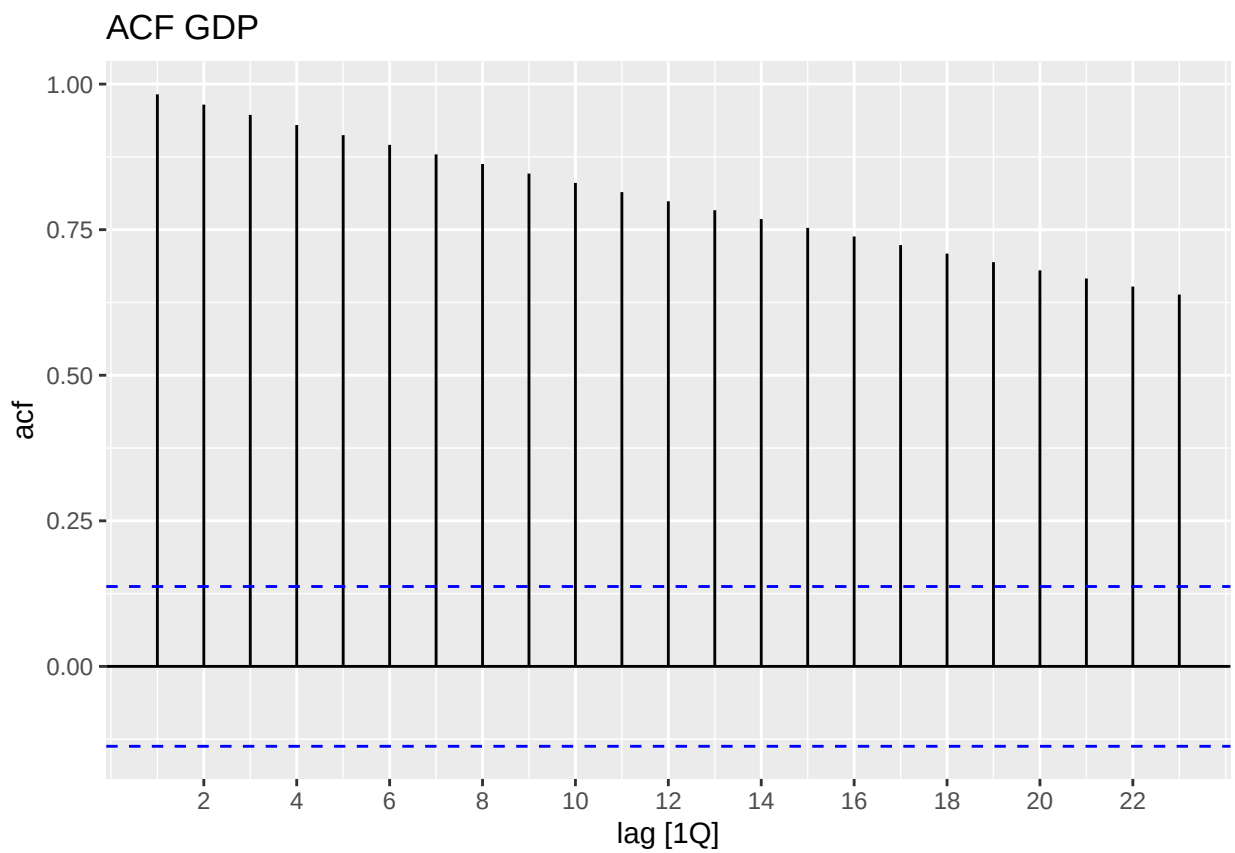
1. Create two plots, one for the GDP series and one for its autocorrelation. What are the relevant features of the data? Can you confirm them from the autocorrelation function? Would it make sense to transform the data? Why?

Solution

```
data("USMacroG", package = "AER")
Macro <- USMacroG %>% as_tsibble(pivot_longer = FALSE)

plot.1 <- Macro %>%
  select(gdp) %>%
    autoplot(gdp) + labs(title = "GDP")

print(plot.1)
```




```
plot.2 <- Macro %>%
  select(gdp) %>%
  ACF(gdp) %>%
    autoplot() + labs(title = "ACF GDP")

print(plot.2)
```



US GDP has a clear increasing trend, with few major fluctuations. It does not show seasonality. The shape of the series looks mildly exponential. This suggests to transform the data with taking logs or with the Box-Cox procedure.

The autocorrelation function confirms that the series has a trend, with all lags significant. The autocorrelation function does not detect any sign of seasonal or cyclical fluctuations, but these can be masked by the trend.

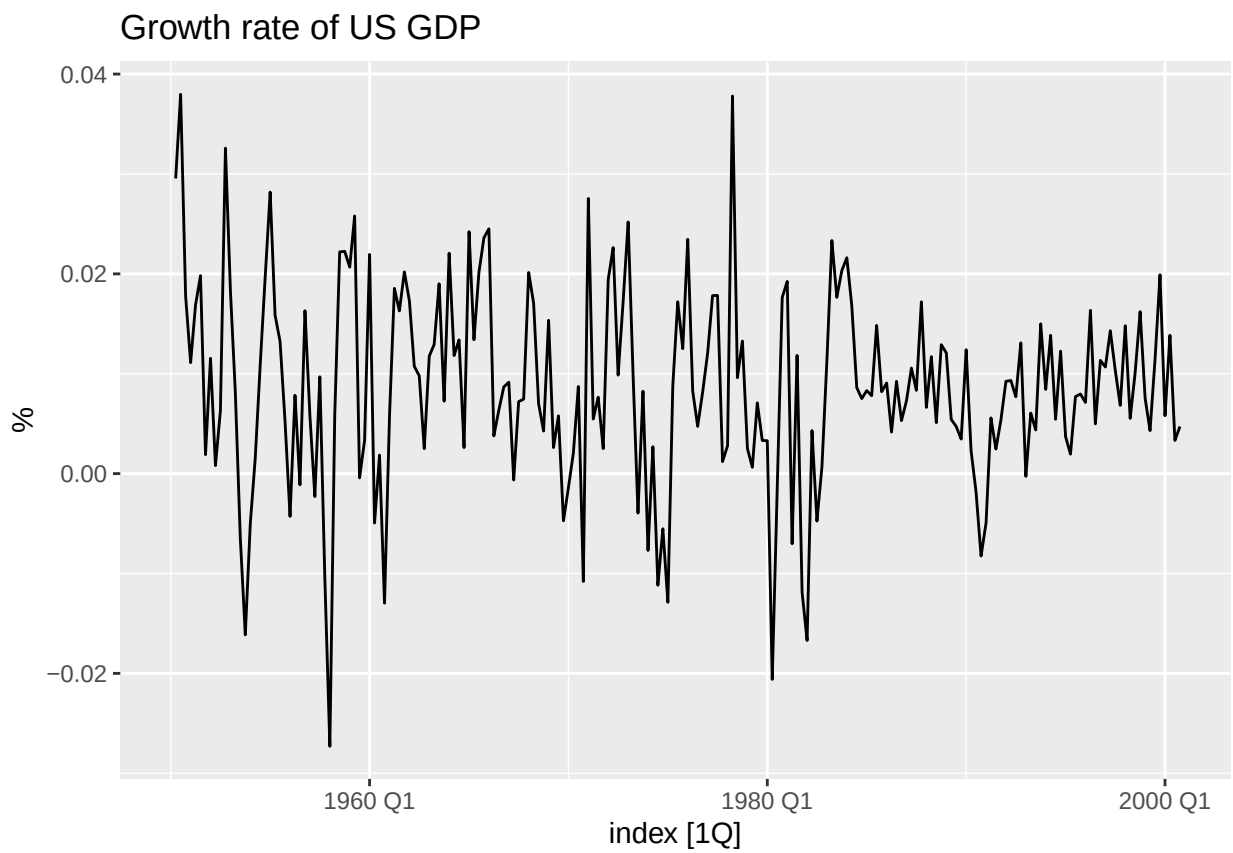
2. Transform the series to create a series for the growth rate of GDP in US. Plot the growth series and its autocorrelation function and comment it. Are the data still trending? Do you see any sign of seasonality or cycle? Perform a statistical test to verify whether data are autocorrelated or not.

Solution

To transform the data, I take the logs and then I need to differentiate them

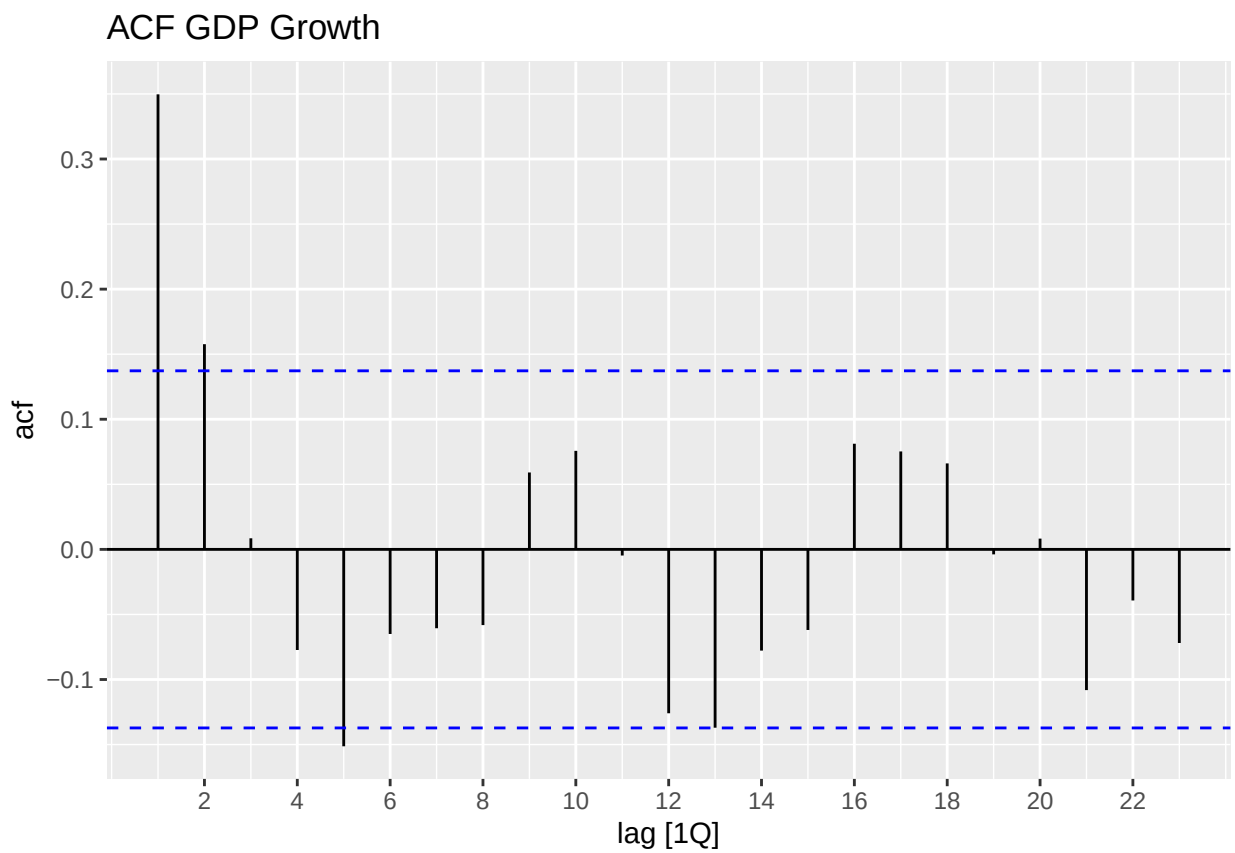
```
Macro %>%
  mutate(growth.gdp = log(gdp) - lag(log(gdp))) -> Macro

Macro %>%autoplot(growth.gdp) +
labs(title = "Growth rate of US GDP") +
ylab("%")
```

```
## Warning: Removed 1 row containing missing values (`geom_line()`).
```



The series do not show a trend anymore. The data seem to randomly fluctuate around a constant mean.

```
Macro %>%
  select(growth.gdp) %>%
  ACF(growth.gdp) %>%
    autoplot() + labs(title = "ACF GDP Growth")
```



The autocorrelation function confirms that the trend is not present anymore. No sign of seasonality. One lag is strongly significant and two more lags *second and fifth are moderately significant, but it is not enough to clearly conclude that there is autocorrelation.

```
Macro %>%
features(growth.gdp, box_pierce, lag = 10, dof = 0)
```

```
## # A tibble: 1 x 2
##   bp_stat bp_pvalue
##     <dbl>     <dbl>
## 1    39.9 0.0000176
```

The Box-Pierce test rejects the null of no-autocorrelation at 1%. Therefore, the evidence supports that the growth rate of US GDP is autocorrelated.

#Question 3

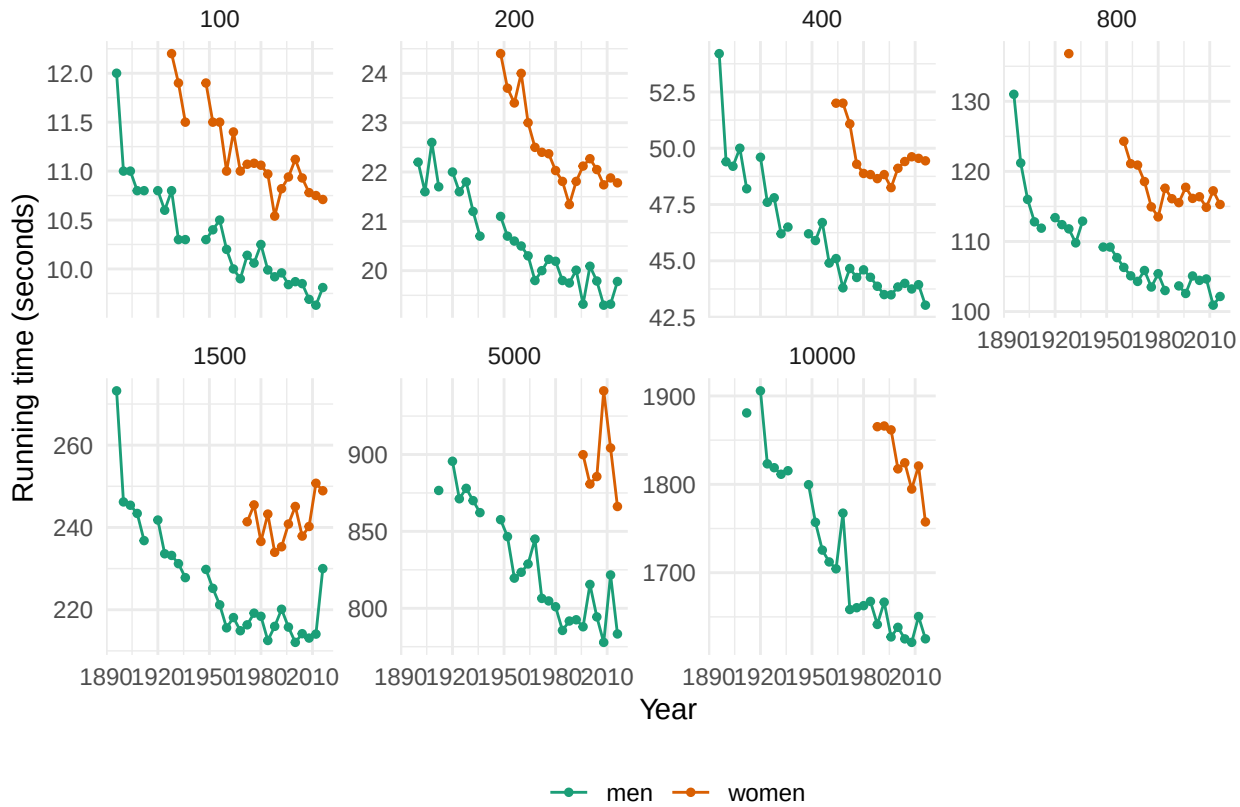
Data set `olympic_running` contains the winning times (in seconds) in each Olympic Games sprint, middle-distance and long-distance track events from 1896 to 2016.

a. Plot the winning time against the year. Describe the main features of the plot.

```
olympic_running %>%
  ggplot(aes(x = Year, y = Time, colour = Sex)) +
  geom_line() +
  geom_point(size = 1) +
```

```r
  facet_wrap(~Length, scales = "free_y", nrow = 2) +
  theme_minimal() +
  scale_color_brewer(palette = "Dark2") +
  theme(legend.position = "bottom", legend.title = element_blank()) +
  labs(y = "Running time (seconds)")
```

```
## Warning: Removed 31 rows containing missing values (`geom_point()`).
```



The running times are generally decreasing as time progresses (although the rate of this decline is slowing down in recent olympics). There are some missing values in the data corresponding to the World Wars (in which the Olympic Games were not held).

b. Fit a regression line to the data. Obviously the winning times have been decreasing, but at what *average* rate per year?

```r
fit <- olympic_running %>%
  model(TSLM(Time ~ trend()))
tidy(fit)
```

```
## # A tibble: 28 x 8
##    Length Sex   .model                term  estimate std.error statistic  p.value
##     <int> <chr> <chr>                 <chr>    <dbl>     <dbl>     <dbl>    <dbl>
##  1    100 men   TSLM(Time ~ trend()) (Int~  11.1      0.0909     123.   1.86e-37
##  2    100 men   TSLM(Time ~ trend()) tren~  -0.0504   0.00479    -10.5  7.24e-11
```

```
## 3    100 women TSLM(Time ~ trend()) (Int~  12.4      0.163       76.0  4.58e-25
## 4    100 women TSLM(Time ~ trend()) tren~  -0.0567   0.00749     -7.58 3.72e- 7
## 5    200 men   TSLM(Time ~ trend()) (Int~  22.3      0.140      159.   4.06e-39
## 6    200 men   TSLM(Time ~ trend()) tren~  -0.0995   0.00725    -13.7  3.80e-13
## 7    200 women TSLM(Time ~ trend()) (Int~  25.5      0.525       48.6  8.34e-19
## 8    200 women TSLM(Time ~ trend()) tren~  -0.135    0.0227      -5.92 2.17e- 5
## 9    400 men   TSLM(Time ~ trend()) (Int~  50.3      0.445      113.   1.53e-36
## 10   400 men   TSLM(Time ~ trend()) tren~  -0.258    0.0235     -11.0  2.75e-11
## # i 18 more rows
```

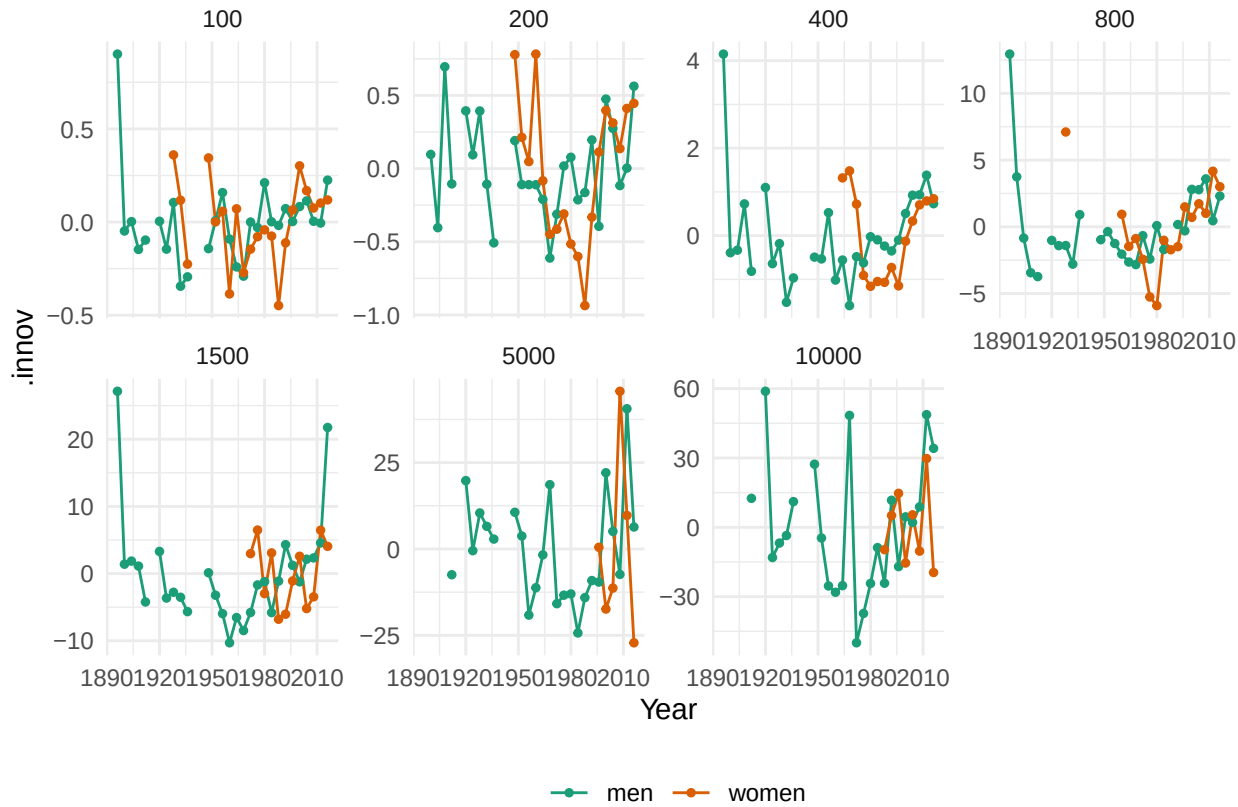
```r
tidy(fit) %>%
  filter(term == "trend()") %>%
  glue::glue_data("The {Sex}'s {Length} running time has been {ifelse(estimate<0, 'decreasing', 'increas
                  by an average of {abs(round(estimate/4, 3))} seconds each year.<br>")
```

The men's 100 running time has been decreasing by an average of 0.013 seconds each year. The women's 100 running time has been decreasing by an average of 0.014 seconds each year. The men's 200 running time has been decreasing by an average of 0.025 seconds each year. The women's 200 running time has been decreasing by an average of 0.034 seconds each year. The men's 400 running time has been decreasing by an average of 0.065 seconds each year. The women's 400 running time has been decreasing by an average of 0.04 seconds each year. The men's 800 running time has been decreasing by an average of 0.152 seconds each year. The women's 800 running time has been decreasing by an average of 0.198 seconds each year. The men's 1500 running time has been decreasing by an average of 0.315 seconds each year. The women's 1500 running time has been increasing by an average of 0.147 seconds each year. The men's 5000 running time has been decreasing by an average of 1.03 seconds each year. The women's 5000 running time has been decreasing by an average of 0.303 seconds each year. The men's 10000 running time has been decreasing by an average of 2.666 seconds each year. The women's 10000 running time has been decreasing by an average of 3.496 seconds each year.

c. Plot the residuals against the year. What does this indicate about the suitability of the fitted line?

```r
augment(fit) %>%
  ggplot(aes(x = Year, y = .innov, colour = Sex)) +
  geom_line() +
  geom_point(size = 1) +
  facet_wrap(~Length, scales = "free_y", nrow = 2) +
  theme_minimal() +
  scale_color_brewer(palette = "Dark2") +
  theme(legend.position = "bottom", legend.title = element_blank())
```

```
## Warning: Removed 31 rows containing missing values (`geom_point()`).
```



It doesn't seem that the linear trend is appropriate for this data.

d. Predict the winning time for each race in the 2020 Olympics. Give a prediction interval for your forecasts. What assumptions have you made in these calculations?

```
fit %>%
  forecast(h = 1) %>%
  mutate(PI = hilo(Time, 95)) %>%
  select(-.model)
```

```
## # A fable: 14 x 6 [4Y]
## # Key:     Length, Sex [14]
##    Length Sex    Year          Time   .mean                          PI
##     <int> <chr> <dbl>        <dist>   <dbl>                       <hilo>
##  1    100 men    2020 N(9.5, 0.061)    9.53 [   9.049446,   10.01957]95
##  2    100 women  2020  N(11, 0.059)   10.5  [  10.056992,   11.01183]95
##  3    200 men    2020   N(19, 0.13)   19.1  [  18.423087,   19.81179]95
##  4    200 women  2020   N(21, 0.31)   21.2  [  20.106551,   22.29293]95
##  5    400 men    2020    N(42, 1.5)   42.0  [  39.668596,   44.41603]95
##  6    400 women  2020    N(48, 1.4)   48.4  [  46.123295,   50.74923]95
##  7    800 men    2020     N(99, 13)   99.2  [  92.137378,  106.33802]95
##  8    800 women  2020    N(111, 14)  111.   [ 104.105124,  118.86174]95
##  9   1500 men    2020    N(207, 74)  207.   [ 190.091940,  223.90462]95
## 10   1500 women  2020    N(245, 35)  245.   [ 233.786710,  257.12632]95
## 11   5000 men    2020   N(773, 285)  773.   [ 739.735488,  805.91363]95
## 12   5000 women  2020  N(892, 1562)  892.   [ 814.664722,  969.56461]95
```

```
## 13  10000 men    2020  N(1580, 970) 1580.   [1519.300480, 1641.40196]95
## 14  10000 women  2020  N(1763, 530) 1763.   [1717.895285, 1808.13543]95
```

Using a linear trend we assume that winning times will decrease in a linear fashion which is unrealistic for running times. As we saw from the residual plots above there are mainly large positive residuals for the last few years, indicating that the decreases in the winning times are not linear. We also assume that the residuals are normally distributed.